

# IBM University Engagement Project Submission

*“Exploring the Power of Agentic AI with IBM Granite and LangFlow”*

# Project Tile – Brain Route AI

also called LEARN MATE

**Domain of Project – AI, Agentic AI, Education Technology (EdTech)**

| Student Name                      | Recent Passport Photo                                                               | Email ID                                                                                                                                                                                   | Mobile number [WhatsApp] |
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# Problem Statement -

## Problem Statement 12 :Agentic AI for Personalized Course Pathways

### The Challenge

Students often struggle to identify the right learning path that aligns with their interests and long-term goals due to the overwhelming number of online courses and a lack of personalized guidance. **Brain Route AI** aims to solve this by acting as an Agentic AI coach that interacts with students, understands their interests (like Frontend Development, Cybersecurity, UI/UX Design, etc.), assesses their current skill level, and dynamically builds a personalized course roadmap that adapts over time based on progress and preferences

### The Objective

Build an AI-powered personalized course pathway assistant called **Brain Route AI** using IBM Cloud, Prompt Lab, IBM watsonx Orchestrate ,and IBM Granite Models.

The system should:

- Recommend personalized learning pathways based on student interests and career goals
- Analyze skill levels and identify missing competencies
- Suggest relevant courses, certifications, and learning resources
- Adapt learning roadmaps based on progress and preferences
- Provide career-focused recommendations for industry readiness
- Help students stay motivated with progress tracking and milestone visualization

## Technology Used

**Tech Stack (Mandatory)** Use of IBM Cloud Lite services / IBM Granite is mandatory

I used:

1. IBM Cloud lite services
2. IBM watsonx Orchestrate
3. Prompt Lab
4. IBM Granite Models
5. IBM watsonx.ai Runtime

**IBM Cloud + Prompt Lab + IBM Granite Models**

**IBM CLOUD + IBM watsonx Orchestrate +IBM watsonx.ai Studio +IBM watsonx.ai Runtime**

# Proposed Solution -

## **Proposed Solution: AI-Powered Nutrition Agent**

Brain Route AI is an intelligent Agentic AI-based learning assistant designed to help students discover the right educational and career pathway according to their interests and goals.

The system uses IBM Granite models through Prompt Lab to understand student preferences, analyze skill levels, and generate personalized learning roadmaps.

The solution works through multiple AI agents collaborating together to deliver adaptive and intelligent learning guidance. The system can recommend courses, certifications, projects, and skill-building activities based on domains such as Frontend Development, UI/UX Design, Cybersecurity, Data Science, AI/ML, and Cloud Computing.

The platform continuously adapts recommendations based on user progress, completed courses, and changing interests. This creates a dynamic and personalized learning experience similar to having a real AI mentor.

The solution also integrates RAG-based retrieval to fetch accurate and relevant course information from trusted learning platforms and educational resources

Overall, Brain Route AI acts as a smart digital career mentor that helps students make informed learning decisions and achieve their long-term goals effectively.

## **Technology Used**

### **IBM Cloud**

Provides a secure and scalable cloud environment for running AI-powered educational solutions and deploying intelligent services.

### **Watsonx Orchestrate**

Used for Agentic AI workflow Conversational interaction AI orchestration User guidance

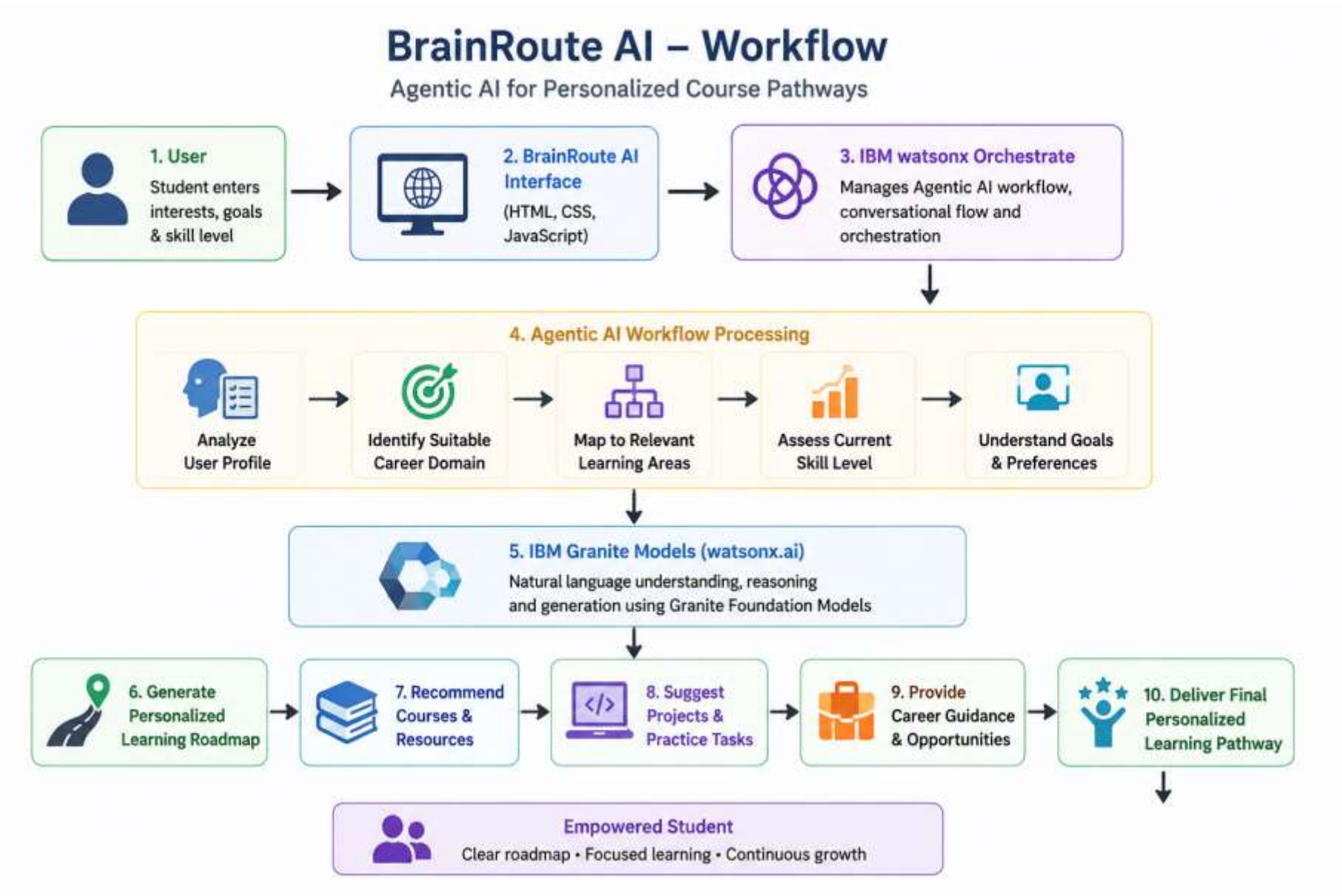
### **IBM Prompt Lab**

Used for prompt engineering, testing prompts, refining AI responses, and generating personalized learning recommendations.

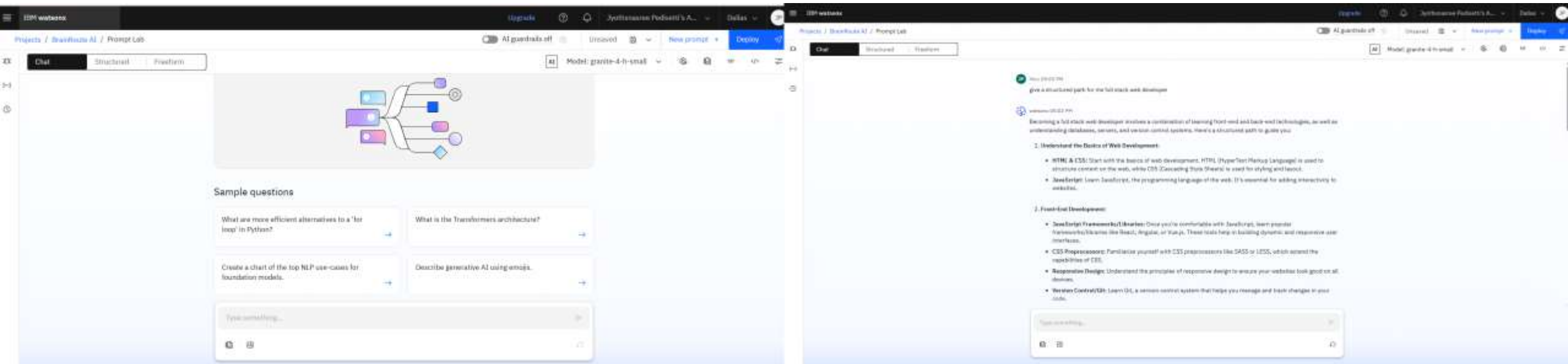
### **IBM Granite Models**

Used for reasoning, roadmap generation, personalized recommendations, summarization, and intelligent conversational guidance.

WORKFLOW



# WORKING SCREENSHOTS: Prompt lab and watsonx Orchestrate



IBM Cloud

Resource list

| Name                        | Group   | Location          | Product            | Status | Tags |
|-----------------------------|---------|-------------------|--------------------|--------|------|
| AI / Machine Learning (2)   |         |                   |                    |        |      |
| watsonx.ai Runtime-v1       | Default | Dallas (us-south) | watsonx.ai Runtime | Active | 1    |
| watsonx.ai Studio-ng        | Default | Dallas (us-south) | watsonx.ai Studio  | Active | 1    |
| Analytics (0)               |         |                   |                    |        |      |
| Application development (0) |         |                   |                    |        |      |
| Asset management (0)        |         |                   |                    |        |      |
| Blockchain (0)              |         |                   |                    |        |      |
| Business automation (0)     |         |                   |                    |        |      |
| Compute (0)                 |         |                   |                    |        |      |



The image displays the BrainRoute AI website and its interactive chat interface. The website header shows the URL <http://127.0.0.1:5506/orchestrator/index.html>. The main heading is "BrainRoute AI", followed by the description: "AI-powered personalized learning assistant developed using IBM watsonx Orchestrator and IBM Granite models. It generates personalized career pathways, skill roadmaps, course recommendations, and project guidance for students." A prominent blue button labeled "Open Live AI Demo" is centered below the text.

The bottom portion of the image shows the chat interface. On the left, a sidebar contains navigation links: "Profile", "Knowledge", "Toolkit", "Behavior", and "Channels". The main chat area on the right features a header with "Brain Route AI" and a status "Offline". The chat content begins with a greeting from the AI: "Hello! I'm Brain Route AI, your personalized learning and career pathway assistant." Below this, there are three interactive buttons: "Recommended career based on my skills", "Create a beginner learning pathway for Data Science", and "Help me choose between AI ML and Cybersecurity". The interface also includes a "Quick start prompts" section with a "Add prompt" button and a "Recommend courses based on my skills" option at the bottom.



The screenshot displays the IBM Watsonx Orchestrate web interface. The top navigation bar includes the IBM logo, the text "IBM watsonx Orchestrate", and links for "English", "Contact Sales", "Buy Now", and a "30 trial days left" badge. On the left sidebar, there are sections for "Agents" (listing "Brain Route AI") and "Chats" (showing a recent chat titled "Recommend courses based on my sk..."). The main chat area shows a conversation with "Brain Route AI" at 6:32 PM. The user's message is "Recommend courses based on my sk...". The AI's response is a detailed learning plan that includes two numbered questions: "1. What areas of technology interest you the most?" and "2. How would you describe your experience right now?". The AI provides example answers for each, such as "Front-end Development, Back-end Development, UI/UX Design" for the first question, and "Beginner", "Intermediate", and "Advanced" levels for the second. At the bottom, there is a text input field with the placeholder "Type something..." and a send button.

IBM watsonx Orchestrate English Contact Sales Buy Now 30 trial days left

Agents

Brain Route AI

Chats

Recommend courses based on my sk...

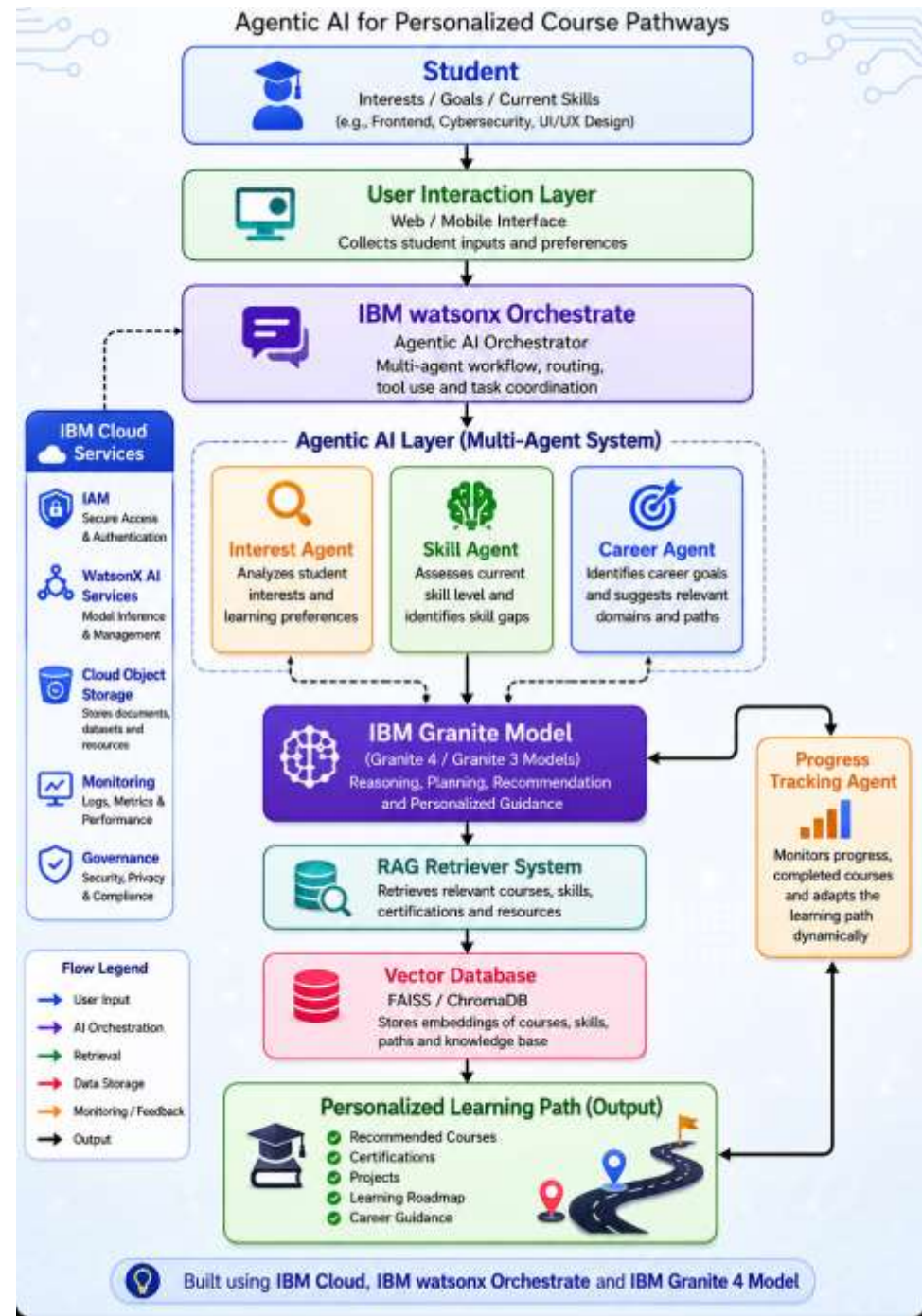
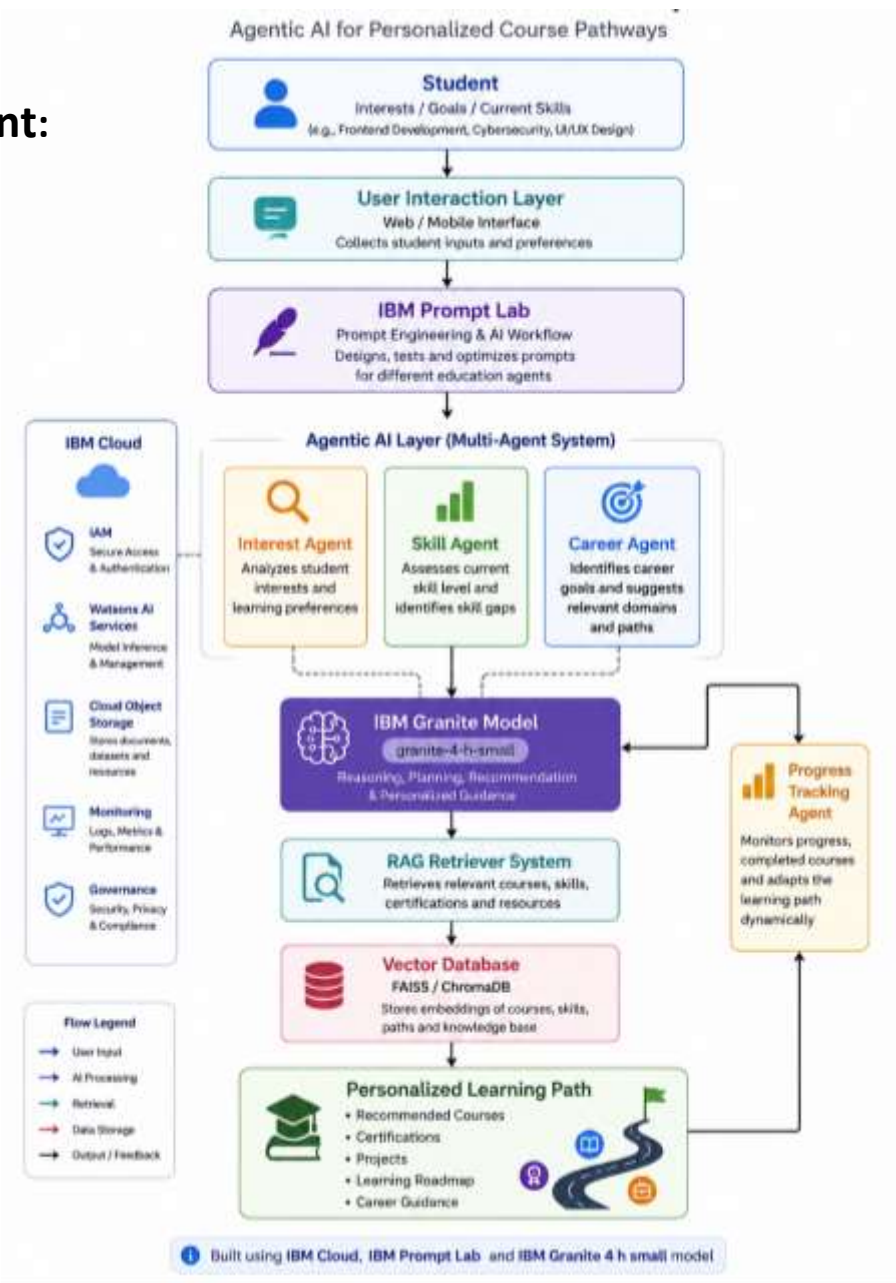
Brain Route AI 6:32 PM

Sure! I'd love to help you map out the perfect learning plan. To recommend courses that match **your current skill level** and **career goals**, could you tell me a bit more about yourself?

1. **What areas of technology interest you the most?**  
(e.g., Front-end Development, Back-end Development, UI/UX Design, Cybersecurity, AI/ML, Data Science, Cloud Computing, Mobile App Development)
2. **How would you describe your experience right now?**
  - **Beginner** – just starting, little to no hands-on experience
  - **Intermediate** – comfortable with basics, have built a few projects or taken an introductory course
  - **Advanced** – strong foundation, several projects or work experience, looking to specialize or deepen expertise

Type something...

Architecture Blueprint:



# Role of Agentic AI in the solution

## Role of Agentic AI in this BRAIN ROUTE AI :

Agentic AI enables Brain Route AI to behave like an intelligent autonomous career mentor instead of a basic chatbot.

The system uses multiple AI agents that collaborate together to deliver personalized educational guidance. Each agent performs a specialized task such as understanding user interests, recommending courses, generating roadmaps, and tracking progress.

The AI system continuously adapts based on user interactions and learning progress, making recommendations more accurate and personalized over time.

Agentic AI also helps in decision-making by analyzing multiple factors including skill levels, learning pace, career goals, and industry trends.

This creates a proactive, adaptive, and goal-driven learning ecosystem that helps students stay focused and career-ready.



# Technology Used

- **IBM Granite model** - IBM-granite 4-h small is used For natural language understanding, reasoning, and personalization
- **IBM watsonx Orchestrate**
- **IBM Cloud** – It provides the infrastructure to build, deploy, and scale the Brain Route AI efficiently. IBM Cloud enables a reliable, secure, and scalable environment for deploying the Agent.
- **RAG**– RAG to fetch real data first, then generate intelligent answers, making the AGENTIC AI COUCH more trustworthy and effective.
- **IBM Watsonx.ai** - For model deployment, embeddings, and AI governance

# Novelty and Uniqueness

## **Personalized Learning Roadmaps**

Generates adaptive course pathways according to each student's goals and skill levels.

## **Dynamic Recommendation System**

Continuously updates recommendations based on progress and preferences.

## **RAG-Based Retrieval**

Combines retrieval with AI generation to provide accurate and trustworthy learning suggestions.

## **Career-Focused Guidance**

Aligns learning recommendations with industry-required skills and certifications.

## **Scalable and Intelligent**

Built using IBM Cloud and IBM Granite Models for enterprise-grade reliability and scalability.

## **AI Mentor Experience**

Acts like a personalized AI learning coach available anytime for guidance and support

## Git Hub Link

<https://github.com/jyothsna0012/Brain-Route-AI->

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Orchestrate link: <https://us-south.watson-orchestrate.cloud.ibm.com/chat>

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# Future Scope

## **1. Integration with Job Platforms**

The system can connect with LinkedIn, Internshala, and other job platforms to recommend skills according to current industry demands.

## **2. AI Interview Preparation**

Can provide mock interviews, resume analysis, and career preparation guidance using AI.

## **3. Multilingual Learning Support**

- Support for regional languages to make learning accessible to more students.

## **4. Integration with Learning Platforms**

- Can integrate with Coursera, Udemy, edX, and YouTube Learning resources for automated recommendations.

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# Thank You!

Thank you for your time and interest.